Weekly Lesson Plan — Week 11 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon | Week of: October 19, 2026 | Course: Engineering Design & Problem Solving (InSPIRESS)

TEKS / ELPS:

TEKS: §127.406(6) implement a design process to develop a multi-system product or solution; 127.406(1) engineering disciplines and professional responsibility
ELPS: c.1C, c.2C, c.3E, c.4C, c.4F, c.5G

Aim of the Lesson / Objective: *Statement of what you want students to do (skill) and/or know (knowledge). Students will…*

MONDAY OCT 19	TUESDAY OCT 20	WEDNESDAY OCT 21	THURSDAY OCT 22	FRIDAY OCT 23
Students will model a thin-walled part using a thin extrude instead of a shell and apply fillets using selection sets. **ALTERNATE:** Students will write 3–4 paragraphs explaining what a branch of engineering is responsible for, and post photographic evidence of the day's work.	Students will complete the door stop using a parametric extrude to an object and a pattern on path with suppressed instances. **ALTERNATE:** Students will write 3–4 paragraphs explaining what a branch of engineering is responsible for, and post photographic evidence of the day's work.	Students will build a model from primitive solids and blend two shapes together with a fillet. **ALTERNATE:** Students will write 3–4 paragraphs explaining what a branch of engineering is responsible for, and post photographic evidence of the day's work.	Students will finish the bulb with edited appearances and model a threaded base using the coil tool. **ALTERNATE:** Students will write 3–4 paragraphs explaining what a branch of engineering is responsible for, and post photographic evidence of the day's work.	Students will audit their own model files against a checklist and repair the features that failed. **ALTERNATE:** Students will write 3–4 paragraphs explaining what a branch of engineering is responsible for, and post photographic evidence of the day's work.

Employed Language Domain(s) Objective: *How students achieve the objective – Listening, Speaking, Reading, Writing.*

Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They listen to team members' ideas, read technical directions, and write explanations of their designs. Friday they audit their own files against a checklist. Alternate-track students read, write and revise a multi-paragraph study of the branches of engineering.

Assessment of Content Objective: *Instrument used to determine mastery.*

Main project: the model file in each student's own cloud folder, checked against the day's feature list and the timeline; Friday, a folder check of Days 6–9. Alternate assignment (all five days): a daily photo of 3–4 written paragraphs posted to Google Classroom, and a Friday conclusion.

Set (Bell Work): *A motivating action that engages students in the objective. Essential Question:*

MONDAY OCT 19	TUESDAY OCT 20	WEDNESDAY OCT 21	THURSDAY OCT 22	FRIDAY OCT 23
why is a door stop a wedge instead of a block?	your part works but uses twice the plastic it needs - do you redesign it?	a light bulb is a cylinder and a sphere stuck together - name three more products built from simple shapes	why does a light bulb screw in instead of plugging in?	which of your four models would actually survive being manufactured, and why?

Methodology / Steps: *What it takes to teach the objective so 100% can pass the assessment.*

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Step 1: Bell ringer / essential question. Step 2: Bell work: why is a door stop a wedge instead of a block? (10 min) Step 3: Day 8 video: triangle profile, Thin Extrude, fillet selection sets (25 min) Step 4: Thin extrude vs shell - when each one is the right call (8 min) Step 5: Save (Ctrl+S) as Day 8 - Door Stop - Lastname (2 min) **ALTERNATE ASSIGNMENT:** Engineering Branches Day 1: what engineering is, and the main branches - 3 to 4 paragraphs + a photo posted to Classroom.	Step 1: Bell ringer / essential question. Step 2: Bell work (dilemma): your part works but uses twice the plastic it needs - do you redesign it? (10 min) Step 3: Day 8 video: grips sketched and constrained, extrude To Object, Pattern on Path (25 min) Step 4: Suppress the instances that do not fit; check it would actually print (8 min) Step 5: Save (Ctrl+S) - Day 8 in your folder (2 min) **ALTERNATE ASSIGNMENT:** Engineering Branches Day 2: civil and structural - what they are responsible for - 3 to 4 paragraphs + a photo posted.	Step 1: Bell ringer / essential question. Step 2: Bell work: a light bulb is a cylinder and a sphere stuck together - name three more products built from simple shapes (10 min) Step 3: Day 9 video: cylinder primitive, intersect for reference, construction lines, sphere, join (25 min) Step 4: Fillet the joint so the transition reads as one smooth part (8 min) Step 5: Save (Ctrl+S) as Day 9 - LED Bulb - Lastname (2 min) **ALTERNATE ASSIGNMENT:** Engineering Branches Day 3: mechanical and manufacturing - what they are responsible for - 3 to 4 paragraphs + a photo posted.	Step 1: Bell ringer / essential question. Step 2: Bell work: why does a light bulb screw in instead of plugging in? (10 min) Step 3: Day 9 video: LED and plastic appearances, threaded base as a new body, Coil tool (25 min) Step 4: Compare your bulb to a real one - what is missing? (8 min) Step 5: Save (Ctrl+S) - Day 9 in your folder (2 min) **ALTERNATE ASSIGNMENT:** Engineering Branches Day 4: electrical, computer and software - what they are responsible for - 3 to 4 paragraphs + a photo posted.	Step 1: Bell ringer / essential question. Step 2: Bell work: which of your four models would actually survive being manufactured, and why? (10 min) Step 3: Catch-up: finish any of Days 6, 7, 8 or 9 that are not done (20 min) Step 4: Quality pass: open each file, fix the broken feature, rename anything named wrong (10 min) Step 5: Folder check with me: four files in Students > your name (5 min) **ALTERNATE ASSIGNMENT:** Engineering Branches Day 5: which branch fits you - a three-paragraph conclusion, plus all five photos turned in.

Resources: *Textbook, handouts, visuals, laptops.*

Google Classroom, projector, laptops, Autodesk Fusion, the "Learn Autodesk Fusion in 30 Days" 2026 Edition videos (Days 8 and 9), engineering notebook, alternate-assignment handout (branches of engineering).

Re-Teaching Activities: *What you prepare if students do not master the objective.*

Rewind-and-rebuild at the teacher table for the feature that failed; printed step list; one-on-one at the student's screen; peer re-teach inside the seating group. Alternate track: sentence stems and a conference.

Reflection: *Students analyze what they accomplished and how they reached mastery.*

Exit ticket – students state which feature they finished and the one mistake they fixed. Alternate track: students name the branch they wrote about and one thing it is responsible for.

Study Skills: *Skills implemented by students.*

Note-taking, technical vocabulary, following multi-step directions, self-checking against a list, teamwork, written argument.

SPED & 504 Accommodations / Modifications: *Codes.*

Per Individualized Education Plan (IEP) / 504 plan.

ELL Accommodations: *Codes.*

Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context and rehearse with a peer before whole-class response (PN, WT). ONE-ON-ONE: individual teacher explanation on request every period; directions restated and modeled at the student desk (CD, RS, VC).

Notes: *Schedule changes this week.*

Full five-day week. Students finish 30 Days of Fusion Day 8 (3D printable door stop) and Day 9 (LED bulb). Friday is a catch-up and quality pass so every student has Days 6 through 9 saved in their own cloud folder before the next unit. The alternate assignment runs all five days: the branches of engineering and what each one is responsible for.